

pyDMS

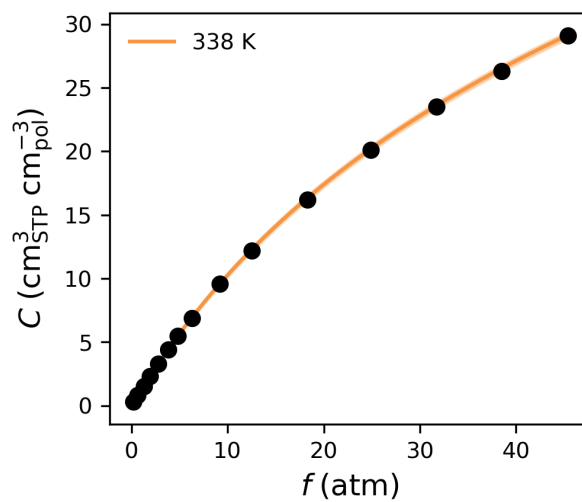
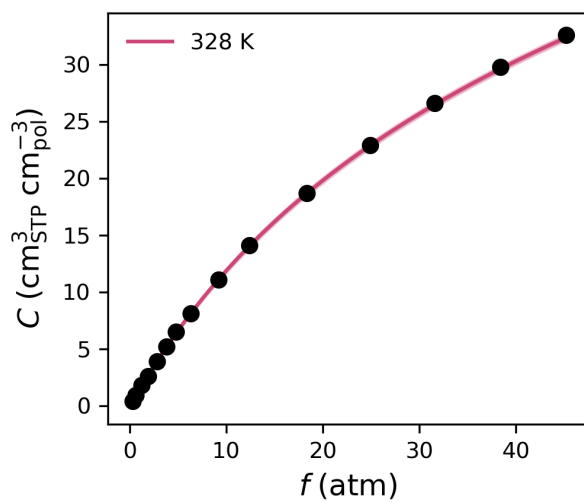
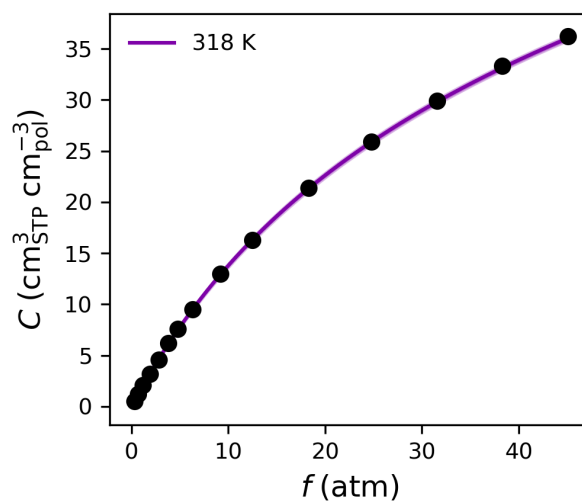
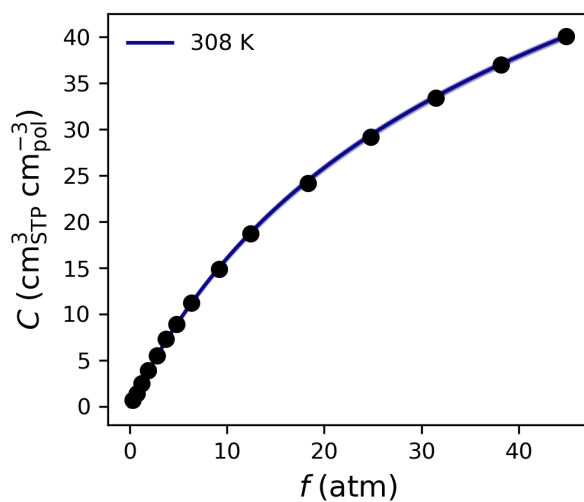
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2025-12-16 17:24:59 Eastern Standard Time

DMS Parameters

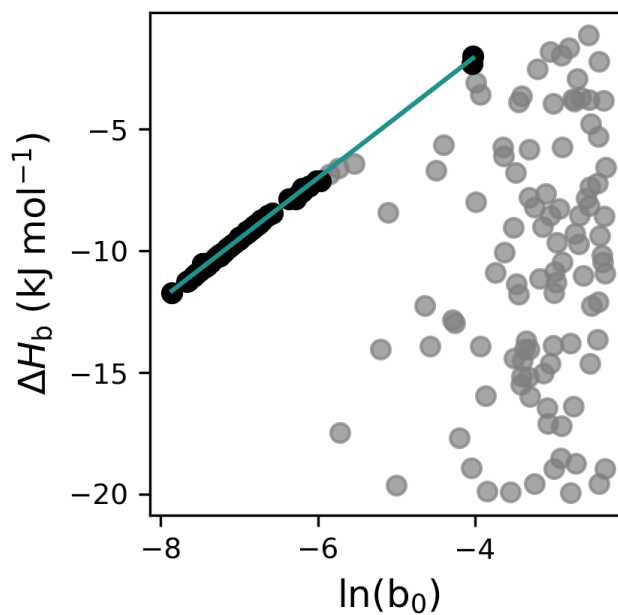
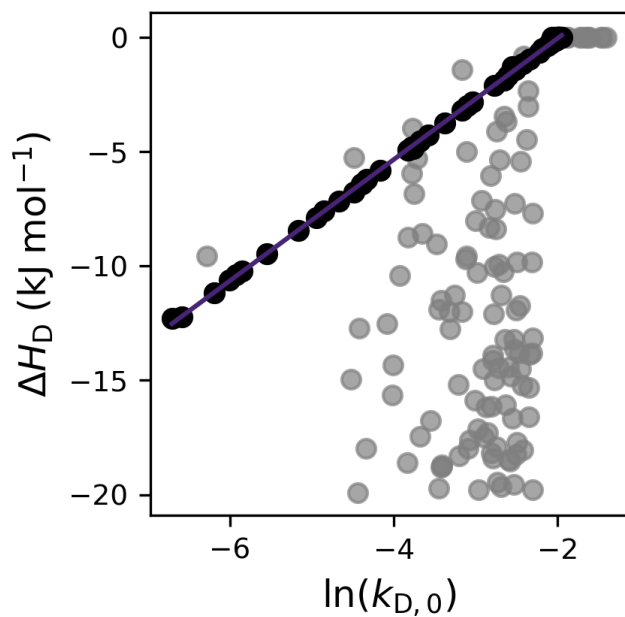
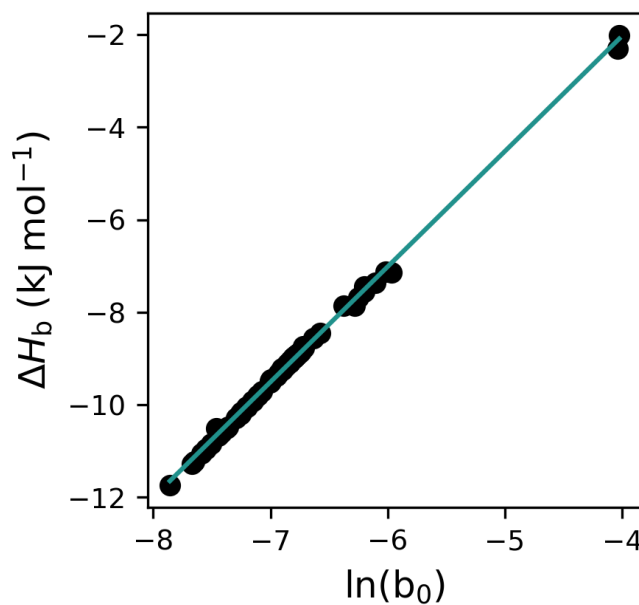
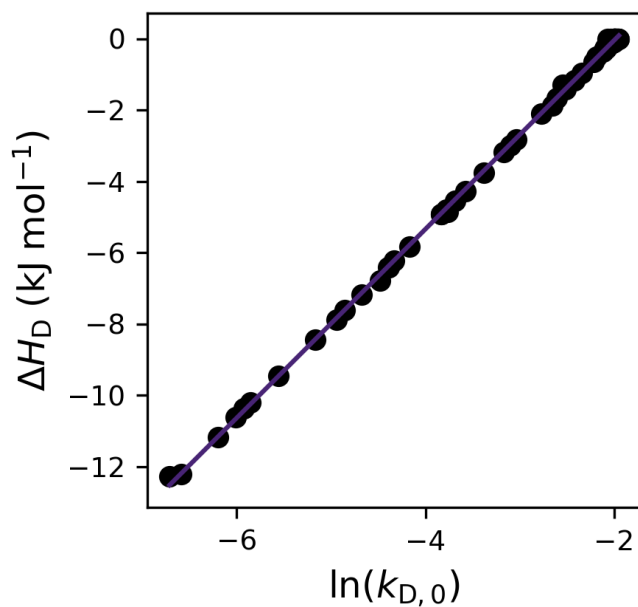
Temperature	Parameter	Result $\pm$ Std. Dev.
308 K	k_D	0.14290 ± 0.00340
	C'_H	53.77360 ± 0.11911
	b	0.03736 ± 0.00157
318 K	k_D	0.13764 ± 0.00119
	C'_H	49.73229 ± 0.15884
	b	0.03325 ± 0.00140
328 K	k_D	0.13288 ± 0.00260
	C'_H	45.96745 ± 0.28968
	b	0.02980 ± 0.00125
338 K	k_D	0.12855 ± 0.00473
	C'_H	42.41078 ± 0.43379
	b	0.02688 ± 0.00113

Sorption Isotherms

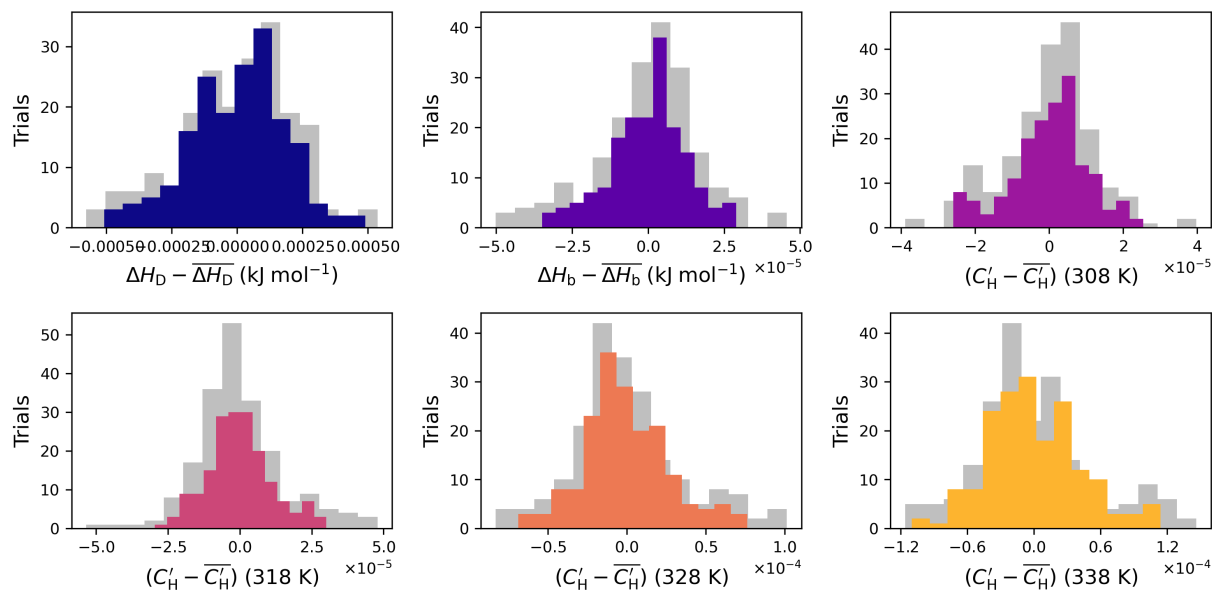


Linear Free Energy Relationships

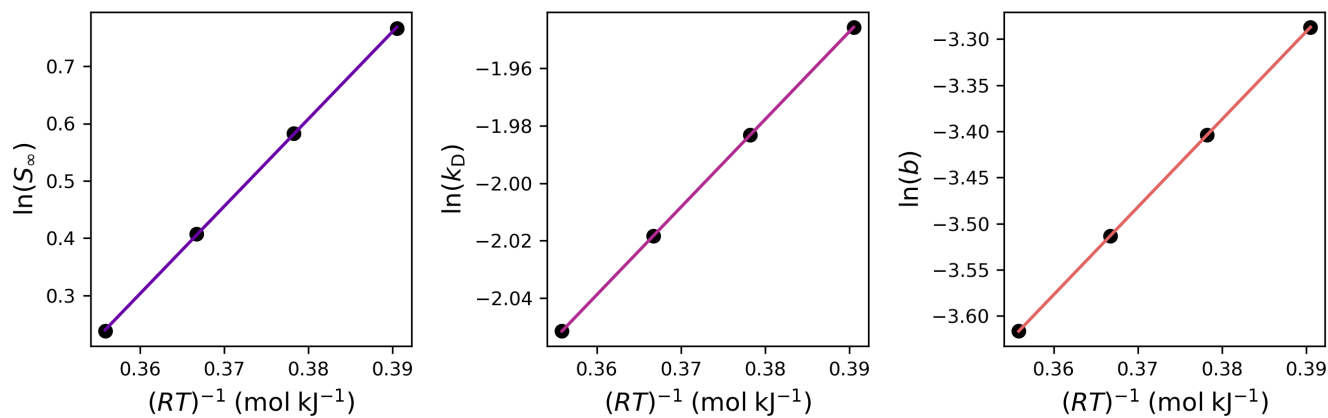
Heat of Henry Sorption (kJ/mol)	$\Delta H_D = 5.2649 + 2.6508 \ln(k_{D,0})$
Heat of Langmuir Sorption (kJ/mol)	$\Delta H_b = 7.952 + 2.4942 \ln(b_0)$



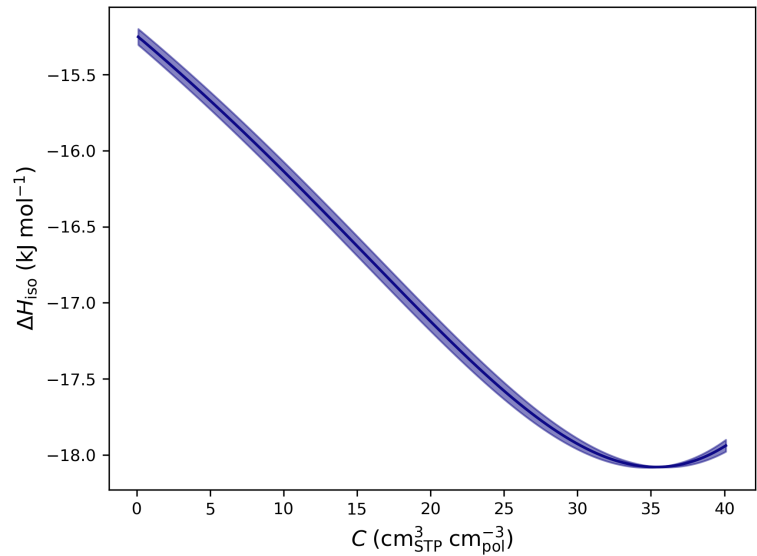
van't Hoff Relationships



Sorption Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	$5.60\text{e-}03 \pm 2.18\text{e-}04$	-15.2444 ± 0.1042
i = Henry	$4.34\text{e-}02 \pm 4.40\text{e-}16$	-3.0531 ± 1.6755
i = Langmuir	$9.14\text{e-}04 \pm 9.78\text{e-}17$	-9.5021 ± 0.0621



Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-20, -1]
ΔH_b Initial Guess Range	[-20, -1]
$k_{D,0}$ Initial Guess Range	[0.01, 0.1]
b_0 Initial Guess Range	[0.001, 0.1]
ΔH_D Solver Bounds	[-30, 0]
ΔH_b Solver Bounds	[-30, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H guess #1	[0 70]
C'_H guess #2	[0 60]
C'_H guess #3	[0 50]
C'_H guess #4	[0 40]
C'_H bounds #1	[0 80]
C'_H bounds #2	[0 70]
C'_H bounds #3	[0 60]
C'_H bounds #4	[0 50]
Trials	200
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-07
xtol (trust-constr)	1e-07
gtol (trust-constr)	1e-07
Seed	None

Questions or Comments?

Please reach out through the pyDMS GitHub page (*INSERT LINK*)

Citation

If you used pyDMS in any presented or published work please cite:

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